

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – IV

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Class: SYBsc IT	Subject: DATA STRUCTURE
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Queue Application:

Write a program to:

a. Implement a queue using an array.

CODE:

```
from array import *

que = array('i', [])

def enqueue(data):
    que.append(data)
    print("Data inserted:", data)

def dequeue():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("Data removed:", que.pop(0))

def peek():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("First element is:", que[0])

def show():
    print("Queue:", que)

enqueue(10)
enqueue(20)
enqueue(30)

show()
peek()

dequeue()
show()
print("IQRA S021")
```

OUTPUT:

```
.>> |
=====
Data inserted: 10
Data inserted: 20
Data inserted: 30
Queue: array('i', [10, 20, 30])
First element is: 10
Data removed: 10
Queue: array('i', [20, 30])
IQRA S021
.>> |
```

b. Simulate a simple queuing system (e.g., customer service queue).

CODE:

```
import time

queue = []

queue.append("DIVYESH")
queue.append("SMIT")
queue.append("DHARMI")
queue.append("SHREYANI")

print("Customers in Queue:")
print(queue)

print("\nProcessing Customers:")

while queue:
    customer = queue.pop(0)
    print(f"Processing: {customer}")
    time.sleep(3)

print("\nQueue is empty.")

print("IQRA S021")
```

OUTPUT:

```
Customers in Queue:
['DIVYESH', 'SMIT', 'DHARMI', 'SHREYANI']

Processing Customers:
Processing: DIVYESH
Processing: SMIT
Processing: DHARMI
Processing: SHREYANI

Queue is empty.
IQRA S021
|
```